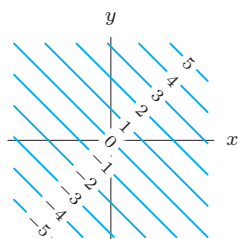


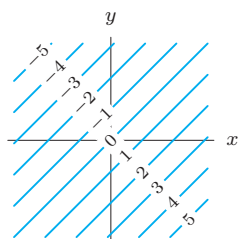
33

T ($^{\circ}\text{C}$)	w (gm/m^3)		
	0.1	0.2	0.3
	10	1300	900
	20	800	800
	30	800	700

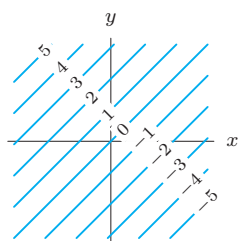
35 (a)



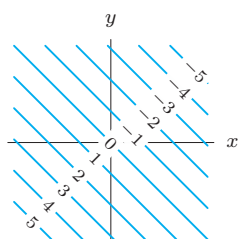
(b)



(c)



(d)



37 $f_y = 0$

39 $f(x, y) = 4x - y$

41 False

43 False

45 True

47 False

49 False

Section 14.2

1 (a) 7.01

(b) 7

3 $f_x(1, 2) = 15$,

$f_y(1, 2) = -5$

5 $\partial z / \partial x = \frac{14x+7}{(x^2+x-y)^{-6}}$

$\partial z / \partial y = -7(x^2 + x - y)^6$

7 $f_x = 0.6/x$

$f_y = 0.4/y$

9 $2xy + 10x^4y$

11 $-2\pi r/T^2$

13 $e^{\sqrt{xy}}(1 + \sqrt{xy}/2)$

15 g

17 $(a + b)/2$

19 $2B/u_0$

21 $2mv/r$

23 Gm_1/r^2

25 $-e^{-x^2/a^2}(a^2 - 2x^2)/a^4$

27 $v_0 + at$

29 $-\pi r^{3/2}/(M\sqrt{GM})$

31 $(15x^2y - 3y^2)\cos(5x^3y - 3xy^2)$

33 $\partial F / \partial L = \frac{3}{2}\sqrt{K/L}$

35 $\epsilon_0 E$

37 $c\gamma a_1 b_1 K^{b_1-1}(a_1 K^{b_1} + a_2 L^{b_2})^{\gamma-1}$

39 13.6

41 (a) 3.3, 2.5

(b) 4.1, 2.1

(c) 4, 2

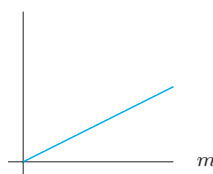
43 (a) $a < 3$

(b) Up

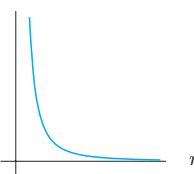
45 (a) $\partial g / \partial m = G/r^2$

$\partial g / \partial r = -2Gm/r^3$

(b) g



g



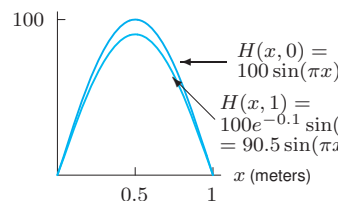
r

47 (a) $c^2((1 - v^2/c^2)^{-1/2} - 1)$; positive

(b) $mv(1 - v^2/c^2)^{-3/2}$; positive

49 (a)

temp ($^{\circ}\text{C}$)



(b) $254.2e^{-0.1t}$ $^{\circ}\text{C/m}$

(c) $-254.2e^{-0.1t}$ $^{\circ}\text{C/m}$

(d) $-10e^{-0.1t} \sin(\pi x)$ $^{\circ}\text{C/sec}$

51 $f(x, y) = x^4y^2 - 3xy^4 + C$

53 Derivative is limit, not a difference quotient
Counterexample: $f(x, y) = x^2 + y^2$

55 $f(x, y) = xy, g(x, y) = xy + y^2$

57 True

59 True

61 False

63 False

65 (d)

Section 14.3

1 $z = ex$

3 $z = 6y - 9$

5 $z = -4 + 2x + 4y$

7 $z = -36x - 24y + 148$

9 $df = y \cos(xy) dx + x \cos(xy) dy$

11 $dz = -e^{-x} \cos(y) dx - e^{-x} \sin(y) dy$

13 $dg = 4 dx$

15 $dP \approx 2.395 dK + 0.008 dL$

17 (a) Dollars/Square foot

(b) Larger plots at same distance \$3/ft² more

(c) Dollars/Foot

(d) Farther from beach but same area \$2/ft less

(e) 998 ft²

19 (b) $f(x, y) \approx$

$0.3345 - 0.33(x - 1) - 0.15(y - 2)$

(c) $f(x, y) \approx 0.3345 -$

$0.3345(x - 1) - 0.1531(y - 2)$

21 376

23 $df = \frac{1}{3}dx + 2dy$

$f(1.04, 1.98) \approx 2.973$

25 136.09 $^{\circ}\text{C}$

27 $P(r, L) \approx$

$80 + 2.5(r - 8) + 0.02(L - 4000)$

$P(r, L) \approx$

$120 + 3.33(r - 8) + 0.02(L - 6000)$

$P(r, L) \approx$

$160 + 3.33(r - 13) + 0.02(L - 7000)$

31 (a) $nRT/(V - nb) - n^2a/V^2$

(b) $\Delta P \approx \frac{nR}{(V_0 - nb)} \Delta T + \frac{(2n^2a/V_0^3 - nRT_0/((V_0 - nb)^2)) \Delta V}{(V_0 - nb)^2}$

33 (a) $d\rho = -\beta\rho dT$

(b) 0.00015, $\beta \approx 0.0005$

35 $-43200\Delta t$

Slow if $\Delta t > 0$; fast if $\Delta t < 0$

37 $z = f(3, 4) + f_x(3, 4)(x - 3) + f_y(3, 4)(y - 4)$

39 Equation not linear

41 sphere of radius 3 centered at the origin

43 True

45 False

47 True

49 False

Section 14.4

1 $(\frac{15}{2}x^4)\vec{i} - (\frac{24}{7}y^5)\vec{j}$

3 $2m\vec{i} + 2n\vec{j}$

5 $\left(\frac{5\alpha}{\sqrt{5\alpha^2 + \beta}}\right)\vec{i} + \left(\frac{1}{2\sqrt{5\alpha^2 + \beta}}\right)\vec{j}$

7 $\nabla z = e^y\vec{i} + e^y(1 + x + y)\vec{j}$